Isotopic Dating of the Newer Volcanics of Victoria, Australia, and Geomagnetic Polarity Epochs

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The Newer Volcanics of Victoria crop out over nearly 25,000 km². K-Ar dating, using the whole-rock technique on 20 samples from 12 widely scattered localities, indicates that activity commenced at least 4.5 m.y. ago and that the main period of eruption was in the Late Pliocene and in the Pleistocene. Some of the basalts contain an appreciable amount of glass or finely crystallized material; from consistency of the data it appears that little or no loss of radiogenic Ar has occurred. The specimens dated were mainly obtained from the same localities that Irving and Green had previously sampled for paleomagnetic measurements. Hence the age data provide additional information on geomagnetic polarity epochs and reversals of the earth's magnetic field. Together with previously published results, the new K-Ar dates strongly suggest that a change from normal to reversed polarity occurred 0.75 ± 0.07 m.y. ago. This polarity change may be the boundary between the Brunhes normal and the Matuyama reversed epochs; alternatively this change may record a short period of reversed polarity (i.e., an event) about 0.81 m.y. ago in an otherwise normal polarity epoch. The results indicate that the age of the boundary between the Matuyama reversed and Gauss normal epochs is poorly controlled at 2.35 ± 0.15 m.y.

INTRODUCTION

The Newer Volcanics of Victoria of Upper Cainozoic age form an extensive plain west of Melbourne covering an area of nearly 25,000 km² (Figure 1). The observed thickness ranges from a meter to more than 70 m. The Newer Volcanics consist mainly of basaltic lavas erupted from volcanic vents, but tuffs also are found. *Edwards* [1938], in a detailed petrological study of the lavas, showed that olivine basalts of alkali affinities are dominant; relatively minor developments of rocks range in composition from limburgite to trachyte.

Much has been written on the age of the Newer Volcanics; geomorphic, paleontologic, and stratigraphic evidence suggests that volcanic activity continued over a considerable period of time. The youthfulness of some lava flows and volcanoes indicates that eruptions occurred in Recent times. Radiocarbon dates on materials closely associated with tuffs provide evidence that activity occurred as recently as 5000 years ago [*Gill*, 1964], although volcanicity appears to be extinct now. The main period of eruption is thought to have occurred during the Pleistocene and Late Pliocene [*Hills*, 1938; *Singleton*, 1941], but views as to the time of initiation of activity range widely from Late [*Singleton*, 1935], Middle [*Hills*, 1938], or Early Pliocene [*Sussmilch*, 1937] to the Late Miocene [*Gill*, 1964] or Miocene [*Walcott*, 1920]. *Singleton* [1941] and *Hills* [1938] pointed out that the lavas appear to be younger than fossiliferous marine sedimentary rocks assigned to the Early Pliocene (Kalinman) of Victoria.

In this study the K-Ar method of isotopic age determination has been applied to a number of samples of the Newer Volcanics. Paleomagnetic measurements by *Irving and Green* [1957*a*, *b*] and *Green and Irving* [1958] showed that some lavas are normally magnetized whereas others have reversed magnetization; if we disregard the sign, the direction of magnetization is parallel to the earth's axis of rotation to within experimental error. This study was initiated because of our interest in obtaining additional data relating to changes in polarity of the geomagnetic field in the geological past.

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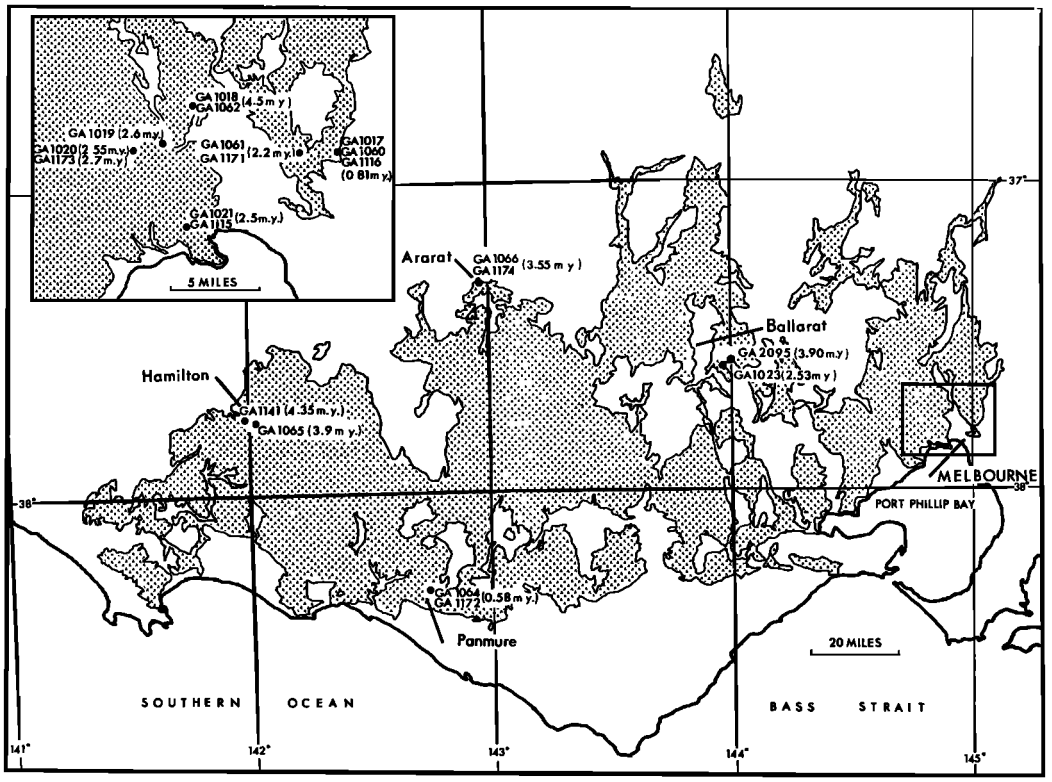


Fig. 1. Map of Victoria showing the distribution of the Newer Volcanics and localities of the samples used in this study.

Whole-Rock Dating of Basalts

The whole-rock technique of K-Ar dating was employed in this study [*McDougall,* 1964]. The sample weight for each Ar extraction usually was about 12 grams, but in some cases up to 35 grams of rock was used. Ar is extracted from the rock by induction heating in a high-vacuum line. To the evolved gases a tracer of Ar^{38} is added, and after purification over hot CuO and Ti the isotopic composition of the Ar is measured in a Reynolds mass spectrometer by the static method [*McDougall,* 1966]. Replicate Ar measurements were made in many cases; Table 1 shows that duplicates agree to within 5% and generally to within 3%. Replicate measurements of K were made flame photometrically by the method of *Cooper* [1963]; agreement to better than 1% was found in each case. In this study the accuracy of the calculated ages is believed to be about ±4% at the 95% confidence level, but for those samples for which the atmospheric Ar contamination exceeds 85% the uncertainty in the measured age may be as great as 10%. Constants used in the calculations are $\lambda_e = 0.585 \times 10^{-10}$ yr^{-1}, $\lambda_\beta = 4.72 \times 10^{-10}$ yr^{-1}, and $K^{40} = 1.19 \times 10^{-2}$ atom %.

The lavas used in this study were all olivine basalts. Phenocrysts of olivine occur in a groundmass of plagioclase, clinopyroxene, olivine, iron oxide, and variable amounts (0 to 15%) of intersertal glass or very fine-grained, imperfectly crystallized feldspathic material. A brief petrographic description of each sample is given in the appendix.

Whole-rock samples yield reliable dates by the K-Ar method if they are holocrystalline and free of alteration [*McDougall,* 1964, 1966; *Dalrymple,* 1963]; otherwise the measured ages may be too young because of leakage of radiogenic Ar. In many of the Victorian basalts

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TABLE 1. Potassium-Argon Dates of Newer Volcanics of Victoria

Sample No.	K Analyses, wt. %		$Ar^{40*}/K^{40} \times 10^{-4}$	Air Correction, %	Calculated Age, m.y.	Polarity
GA 1064	1.125	1.126	0.325	71.6	0.56	N
	1.127					
GA 1172	1.094	1.093	0.342	75.8	0.586	
	1.091					
			0.474	44.9	0.81	
GA 1017	1.092	1.094	0.471	51.9	0.81	
	1.096		0.483	48.2	0.83	
GA 1060	1.128	1.130	0.475	88.8	0.81	R
	1.132		0.448	57.6	0.77	
GA 1116	1.127	1.124	0.471	53.7	0.81	
	1.121		0.479	40.5	0.82	
GA 1061	0.720	0.721	1.264	47.7	2.16	R
	0.722					
GA 1171	0.812	0.815	1.307	61.3	2.23	
	0.818					
GA 1023	1.306	1.306	1.500	38.2	2.57	N
	1.307		1.458	48.5	2.49	
GA 1021	0.743	0.743	1.457	83.2	2.49	N
	0.743					
GA 1115	0.728	0.730	1.464	90.1	2.50	
	0.732					
GA 1019	0.759	0.760	1.542	79.8	2.63	N
	0.760					
GA 1020	0.747	0.750	1.49	83.7	2.55	N
	0.752					
GA 1173	0.662	0.664	1.61	90.4	2.74	
	0.666					
GA 1066	0.582	0.583	2.11	86.9	3.59	R
	0.584					
GA 1174	0.646	0.647	2.06	86.1	3.52	
	0.648					
GA 2095	1.560	1.560	2.26	19.7	3.87	R
	1.561		2.30	20.3	3.93	
GA 1065	1.158	1.160	2.31	49.3	3.95	R
	1.162		2.28	66.1	3.88	
GA 1141	0.717	0.717	2.54	33.3	4.35	N
	0.718					
			2.80	19.4	4.68	R
GA 1018	0.970	0.971	2.61	19.0	4.46	
	0.972		2.65	26.7	4.52	
			2.59	25.3	4.42	
GA 1062	0.876	0.877	2.60	55.0	4.45	
	0.878		2.58	24.8	4.52	

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olivine is partly or, in some cases, completely altered to iddingsite. This alteration does not appear to affect the measured age because the K content of olivine is negligible [*Dalrymple,* 1963; *McDougall,* 1964]. In the present study those rocks which showed evidence of alteration of plagioclase and pyroxene, from examination of a thin section, were rejected. We thought that radiogenic Ar might not have been retained quantitatively in the fine-grained or imperfectly crystallized intersertal material, which occurs in many of the basalts, particularly if this material had formed by devitrification of glass. An appreciable amount of K probably occurs in this material. Only those samples in which the intersertal material showed little or no evidence of alteration were dated.

In general, consistency of results is the only means available to check whether leakage of radiogenic Ar has or has not occurred. Hence, for those localities where the samples contained poorly crystallized material, two specimens from different parts of the same lava flow usually were dated. If Ar leakage had taken place, it is unlikely that it would occur to the same degree in each sample. The remarkably good agreement found between the results on different samples from the same lava flow (Table 1) provides a firm basis for believing that little or no Ar loss has occurred.

This agreement is well demonstrated by the results for three samples from the Albion quarry at Alphington, Melbourne. The quarry is about 30 m deep and is cut in a single, thick olivine basalt lava flow. Two samples (GA 1017, GA 1116) from the northern end of the quarry and near the top of the flow, just below the scoriaceous upper zone, are essentially holocrystalline basalts almost completely free of deuteric alteration. However, intersertally there occurs about 2% of isotropic to finely crystalline material. The dates for these two samples agree closely at 0.81 m.y. (Table 1). The third sample (GA 1060) was collected several hundred meters to the south, from the lowest exposed part of the lava. Although petrographically similar to the other samples, the olivine, pyroxene, and plagioclase show incipient chloritization and there is about 10% of imperfectly crystallized, somewhat chloritized, material present intersertally. The alteration appears to be deuteric. The Ar run for this sample has a much higher proportion of atmospheric Ar than the unaltered rocks; this seems to be related to the degree of alteration. The indicated age is identical with that found for the two fresh rocks from the same lava flow (Table 1).

It is concluded from this example and from the other results (Table 1) that the presence of some deuteric alteration does not necessarily cause loss of radiogenic Ar from volcanic rocks which have remained at essentially room temperature since they cooled. In any given case there is no a priori way of knowing whether this is true; to provide confidence in the results it is essential to date at least two samples from each lava flow that shows some alteration.

Two samples that we regarded as petrographically suitable for dating were found to have such a high proportion of air Ar (> 96%) that the data were rejected. Results from a third sample were also ignored because the air Ar content exceeded 90% of the total Ar in the three runs made, and the spread of the indicated ages was 25%. All other measurements are reported in this paper.

Dating Results and Discussion

The analytical data are given in Table 1. Although 19 samples have been dated, only 13 flows are represented because two samples have been analyzed from the same basalt flow in many cases. The samples were obtained from widely scattered localities (see Figure 1), but because of the limited number of rocks dated this study can only be regarded as of a reconnaissance nature. However, it is thought that a fairly representative group of rocks, as regards age, has been dated.

The results show that volcanic activity commenced at least 4.5 m.y. ago. Because of correlation difficulties the age of the Pliocene-Miocene boundary is in doubt; if the Clarendonian is included in the Pliocene, the age of the boundary is about 12 m.y. [*Evernden et al.,* 1961], but if the Clarendonian is regarded as Upper Miocene the boundary is at approximately 7 m.y. [*Funnell,* 1964]. The Pliocene-Pleistocene boundary commonly has been assumed to be about 1 m.y. old; however, *Evernden and*

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Curtis [1965] present evidence that it may be as old as 3 m.y. Funnell indicates that the age of this boundary may lie between 1.5 and 3.5 m.y. Notwithstanding these uncertainties, our results show that the main period of eruption of the Newer Volcanics of Victoria was in the Late Pliocene and in the Pleistocene.

Melbourne area. A relatively large number of samples was collected from the Newer Volcanics over an area of about 180 km² in the vicinity of Melbourne, at the eastern margin of the lava plain (Figure 1). The measured dates range from 0.8 to 4.5 m.y., covering almost the entire age span found for these lavas.

At Alphington, at the extreme eastern edge of outcrop of the Newer Volcanics, three samples from the same quarry gave dates agreeing at 0.81 ± 0.03 m.y., and, as discussed earlier, this date is considered to be close to the true age. Both samples of basalt from the Merri Creek quarry, three miles to the west of Alphington, yielded results that agree closely at 2.20 ± 0.07 m.y. In the field it is not possible to determine the stratigraphic relationships between the two localities.

A number of basalts from the western outskirts of Melbourne in the Essendon-Maribyrnong area have been measured. Fowler's quarry, North Essendon, is cut mainly in a basalt flow more than 20 m thick. Two samples collected about 400 m apart from the south and west walls of the quarry gave reproducible dates agreeing at about 4.50 ± 0.15 m.y. Both samples are free of alteration, with a few per cent of pale brown, unaltered isotropic glass and minor poorly crystallized material filling vesicles.

Extending southward from the Maribyrnong River in the Braybrook, Sunshine, and Newport areas is an extensive, little dissected basalt plain which slopes gently south to sea level at Port Phillip Bay. Within the plain numerous quarries have been operated, exposing one or two lava flows. The basalts are strikingly similar over the whole area; they are dark to light gray, finely vesicular olivine basalts with high but variable amounts of intersertal glass or finely crystallized feldspathic material. Olivine commonly is partly altered to iddingsite. The Newport quarry is about 8 km from the Braybrook and Sunshine quarries, which are about 3 km apart. The dates for four of the five samples agree at 2.55 ± 0.15 m.y. Because of the low proportion of radiogenic Ar (10 to 20%) the uncertainty in each measurement may be as high as 10%. The similarity of the lavas to one another strongly suggests that these four samples are from a single extensive flow or from closely related flows, and this is supported by the agreement between the dates and by the essentially identical K contents. The other basalt (GA 1173) is possibly from a lava flow below that from which sample GA 1020 was obtained, but the field relations are not clear. The older age found for this rock is consistent with this view. These dates all are internally consistent, and, although an appreciable amount of glass or finely crystallized material occurs in each sample, the good agreement found suggests that the ages are valid.

Western Victoria. Results were obtained for samples from a number of widely separated localities in western Victoria (Figure 1). The youngest dates found in this study were for two samples of the same lava flow near Panmure (Table 1). Two lava flows dated were from separate quarries about 1.5 km apart from Dunnstown, near Ballarat. The measured ages are 2.53 ± 0.07 m.y. for GA 1023 and 3.90 ± 0.10 m.y. for GA 2095. Both rocks are fresh holocrystalline basalts and are ideal for dating. Because of lack of outcrop, the relative stratigraphic age of the lavas cannot be determined in the field. At the northwestern extremity of the lava plain in the Ararat area, two samples (GA 1066, GA 1174) from adjacent quarries give dates agreeing at 3.55 ± 0.10 m.y.

It has been recognized on geomorphic and stratigraphic grounds that some of the oldest basalts of the Newer Volcanics crop out in the Hamilton area near the western margin of the lava plain [*Gill,* 1964; *Gibbons and Gill,* 1964]. At Grange Burn, just west of Hamilton, *Gill* [1957, 1964] described basalt overlying a fossil soil developed on the Kalinman limestone of Early Pliocene age. A K-Ar date of 4.35 m.y. was obtained for a sample of the basalt; this result was reported previously [*Turnbull et al.,* 1965]. A further sample from a different lava flow about 1.5 km east of Hamilton gives a reproducible date of 3.9 ± 0.1 m.y. The relation of this lava to that at Grange Burn is unknown.

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Geomagnetic Polarity Epochs

The hypothesis that the geomagnetic field has changed its polarity in the geological past has recently received strong support from combined paleomagnetic and K-Ar dating studies. The pattern of normal and reversed polarity has now been delineated back to about 4 m.y. ago, but considerable uncertainty still exists as to the time at which some of the changes in polarity occurred [*Cox et al.*, 1964, 1965; *McDougall and Tarling*, 1963, 1964]. In addition, evidence is now accumulating that within a given polarity epoch there may be relatively short intervals during which the opposite polarity prevails; an example of this is the Olduvai normal event in the Matuyama reversed epoch [*Cox et al.*, 1964, 1965]. The age measurements on the Victorian Newer Volcanics provide additional data relevant to geomagnetic polarity epochs.

The paleomagnetism of the Newer Volcanics was studied by *Irving and Green* [1957*a*, *b*] and *Green and Irving* [1958]. They sampled from 32 quarry sites and found normal polarity for 13 sites, reversed polarity for 16 sites, and mixed polarity for 3 sites. The tight grouping of the magnetic directions for samples from within a site and between sites, together with the fact that the polarities are exactly opposite, was considered by Irving and Green to be strong evidence that the magnetization was stable and thermoremanent in origin. *Parry* [1960] thermomagnetically analyzed representative samples and found no significant difference in behavior between rocks of normal and reversed polarities. His results showed that the main magnetic phase in these basalts is a titanomagnetite of somewhat variable composition. *Stacey*'s [1960] measurements on several samples of the Newer Volcanics indicate that magnetic anisotropy is less than 2%.

The original paleomagnetic specimens were not available for this study; hence it was necessary to resample in the field. With one possible exception in each of the quarries examined, only a single lava flow is exposed, so that it is reasonable to assume that for each site our specimens were obtained from the same lava flow as Green and Irving's. Petrographic comparison of thin sections cut from Green and Irving's cores and the samples collected for this study provided confirmation of the equivalence of the two sets of samples. As a further check, oriented samples were collected from all but three of the sites dated, and in each case the new paleomagnetic results were in satisfactory agreement with the earlier data.

Selected specimens from the new collection were demagnetized by alternating fields in steps of 50 to 350 oe (peak). The magnetization was found to be stable except for a small but variable component of viscous magnetization that could be removed by a field of 150 oe. The remainder of the samples were then measured after treatment at 150 oe. For those sites for which no additional samples were collected, Green and Irving's specimens were used for the demagnetization experiments. No samples were available from the Ararat quarry, but because of the uniform stability of the other samples studied there appears to be no reason for questioning the polarity given by Green and Irving for this site. The paleomagnetic results are summarized in Table 2 and Figure 2. The data indicate that the NRM is stable and that it is thermoremanent; there is no evidence that self-reversal mechanisms have operated.

Magnetic polarity versus age for the Victorian basalts is given in Table 1 and shown in Figure 3, together with previously published results from elsewhere. In Figure 3 several symbols are used in an attempt to distinguish between results of differing reliability. The present data generally are in satisfactory agreement with the earlier results; however, they provide information on two critical polarity epoch boundaries. In addition, there are two determinations on rocks in the age range 4 to 4.5 m.y.; very few data are available for rocks of this age.

The recently published results for rocks from Réunion [*Chamalaun and McDougall*, 1966] give much-needed detailed coverage in the age range 0 to 0.6 m.y. and amply confirm that normal polarity prevailed during this time. The result at about 0.57 m.y. from Victoria is an additional data point.

The age of the boundary between the Brunhes normal and the Matuyama reversed epochs, following the nomenclature of *Cox et al.* [1964], has commonly been given as 1.00 ± 0.05 m.y. The date assigned was dependent mainly on

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TABLE 2. Magnetic Results for Samples from Dated Sites in the Newer Volcanics of Victoria

Locality	*Green and Irving* [1958] Site No.	Mean Direction		This Study Mean Direction after ac Demagnetization		No. of Samples	Resultant Vector R
Quarry near Panmure	NV 27	348.0	−67.0	337.0	−47.7	1	
Albion quarry, Alphington	NV 15	149.6	+34.9	146.3	+54.0	4	3.939
Merri Creek quarry	NV 12	192.0	+74.0	180.0	+66.6	3	2.988
Disused quarry, Dunnstown				355.0	−54.4	4	3.991
Operating quarry, Newport	NV 3	055.0	−71.0	007.6	−69.0	4	3.819
McGrath's quarry, Braybrook	NV 9	343.0	−58.0	348.0	−57.0	4	3.960
Albion quarry, Sunshine	NV 10	00.3	−61.8	353.1	−59.0	5	4.974
Ararat quarry	NV 28	205.0	+68.0				
	NV 29	210.0	+71.0				
Operating quarry, Dunnstown	NV 18	152.0	+61.0	170.0	+53.0	3	2.944
Menzel's quarry, Hamilton	NV 30	188.0	+50.0	183.0	+41.0	1	
Grange Burn, near Hamilton				006.8	−53.6	1	
Fowler's quarry, Essendon	NV 11	157.0	+54.0	156.0	+53.4	3	2.927

the age of the normally magnetized Bishop tuff, California, and on the age of the reversely magnetized obsidian at Big Pine, California [*Cox et al.*, 1963*a*]. The Bishop tuff was dated at about 0.98 m.y. by *Evernden et al.* [1964] and the Big Pine obsidian was dated at 0.99 m.y. [*Cox et al.* 1963*a*]. *Dalrymple et al.* [1965] showed, however, that the Bishop tuff probably is about 0.7 m.y. old., and *Cox et al.* [1965] and *Doell et al.* [1966] therefore revised their estimate for the age of the boundary between the Brunhes normal and the Matuyama reversed epochs to 0.85 ± 0.15 m.y.

In the present study three samples of basalt from the Alphington quarry, Melbourne, were dated at 0.81 ± 0.03 m.y.; as discussed earlier, the dates are in excellent agreement and are considered to provide an accurate measure of the age. *Green and Irving* [1958] collected 34 oriented samples from this particular quarry and found that the measured paleomagnetic directions are tightly grouped and show consistently reversed polarity. During the course of this work, additional oriented samples were collected and shown to have the same polarity as those of Green and Irving. One of these specimens (GA 1116) was used in the dating. Demagnetization experiments on several samples indicated stability. Further confirmation that at about this time the polarity was reversed is provided by a basalt from France that has reversed polarity and is dated at 0.81 ± 0.03 m.y. (McDougall, unpublished data). Together with the Bishop tuff results, these data suggest that a change in polarity from normal to reversed occurred 0.75 ± 0.07 m.y. ago.

Because of the few available results in the age range 0.7 to 1.0 m.y., it is not known whether this change of polarity can be accepted as the boundary between the Brunhes normal and the Matuyama reversed epochs (Figure 3). The 0.94 ± 0.16 m.y. date on plagioclase from the Devils Postpile basalt, which has normal polarity [*Cox et al.* 1963*b*], is too imprecise to be of much value. Two rocks

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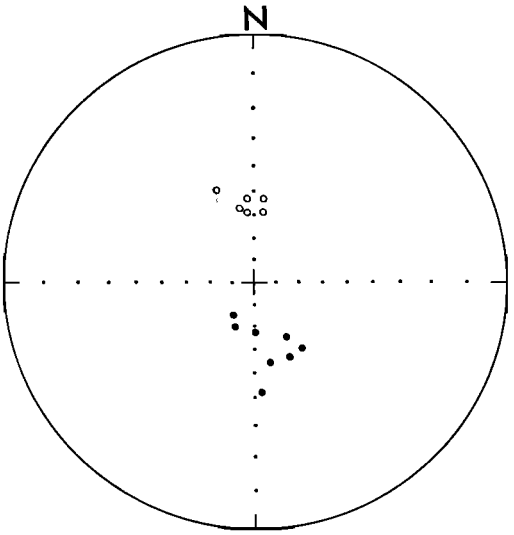


Fig. 2. Stereographic projection of site means after ac demagnetization to 150 oe (peak).

from the Kula series of East Maui, Hawaii, were reported by *McDougall and Tarling* [1963] as having normal polarity and were dated at 0.84 and 0.43 m.y. The samples dated were not measured paleomagnetically, although other rocks from the same formation possessed normal polarity [*Tarling*, 1965]. Because of the uncertainty in polarity arising from the fact that the Kula series was erupted over a relatively long period of time, it is now considered that these data should be rejected; they are not included in Figure 3. *Chamalaun and McDougall* [1966] recorded normal polarity for a basalt dated at 1.01 ± 0.05 m.y. from Réunion; this result, together with others from the same volcano and previously published data, indicates that another change in polarity from normal to reversed occurred 1.00 ± 0.05 m.y. ago. Hence the results suggest that changes in polarity occurred 0.75 ± 0.07 and 1.00 ± 0.05 m.y. ago. One of these changes probably records an event, and the other the boundary between the Brunhes normal and the Matuyama reversed polarity epochs. At present we have too few data to decide which is the event and which is the boundary between the epochs; the choice may prove to be an arbitrary one. The possibility that self-reversal is responsible for some of the apparent difficulties

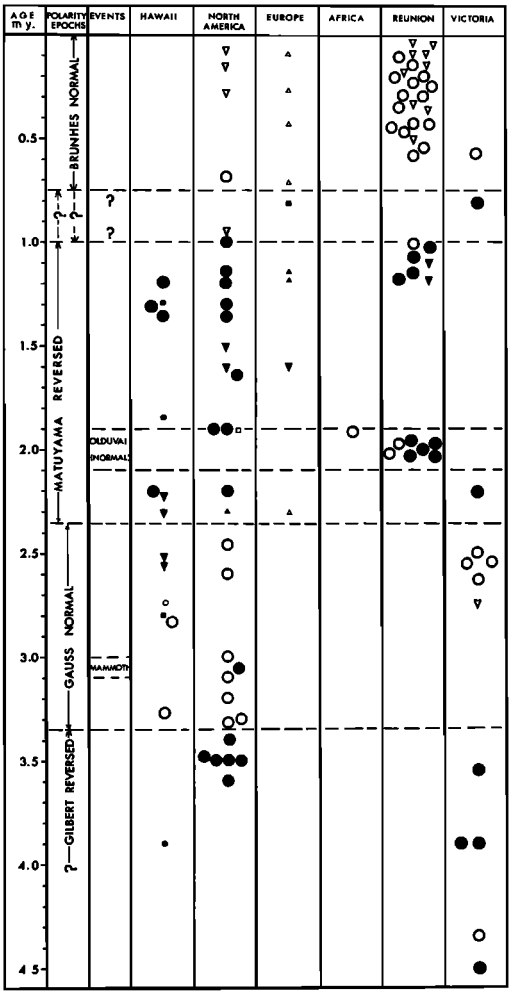


Fig. 3. Diagram of magnetic polarity versus age for the Newer Volcanics of Victoria, together with data from elsewhere. The nomenclature of the epochs and events is that of *Cox et al.* [1964]. An attempt has been made in the diagram to assess the quality of the data (see also *Doell et al.* [1966]). Open symbols, normal polarity; filled symbols, reversed polarity; large circle, data appear to be satisfactory; small circle, date determined for different sample from that on which polarity measurement was made, although generally from the same formation; triangle, polarity determined using hand compass; inverted triangle; uncertainty in age measurement is greater than 10%; small square, detailed information not published. Sources of data: Hawaii, *McDougall and Tarling* [1963], *McDougall* [1964], *Evernden et al.* [1964], *Cox et al.* [1965], this paper. Sierra Nevada, Europe, and Africa, see summary in *Doell et al.* [1966] for complete references. Réunion, *Chamalaun and McDougall* [1966].

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should also be borne in mind. Clearly, additional data must be obtained on rocks whose ages lie within the range 0.7 to 1.0 m.y. in order to distinguish between the alternative interpretations.

A reasonably large number of consistent results is now available for the Matuyama reversed epoch, and the reality of the Olduvai normal event within this epoch appears to have been established [*Cox et al.*, 1965; *Chamalaun and McDougall*, 1966; *Doell et al.*, 1966]. Several results from Victoria are important in that they provide some further control as to the age of the boundary between the Matuyama and Gauss normal polarity epochs. Sample GA 1023 from a quarry near Dunnstown was dated at 2.53 ± 0.07 m.y.; this rock is holocrystalline and completely free of alteration, and hence the age is regarded as reliable. Normal polarity and a direction that was very stable during demagnetization was found for four oriented samples collected from this quarry. The basalts of normal polarity from the Sunshine, Braybrook, and Newport areas are dated at 2.55 ± 0.15 m.y. (three samples) and 2.74 ± 0.20 m.y. All these results clearly indicate that the boundary between the Matuyama reversed and Gauss normal epochs is dated at 2.5 m.y. or less. Taking into account results from elsewhere, but specifically excluding those for which the dates have a large uncertainty (Figure 3), we conclude that the age of the Matuyama-Gauss boundary remains poorly controlled at 2.35 ± 0.15 m.y.

McDougall and Tarling [1963] reported a date of 2.94 m.y. for a lava from the Waianae series of Oahu, Hawaii, and the polarity was given as reversed. The date lies within the period covered by the Gauss normal epoch. Although the sample dated was believed to have been collected from the same lava flow as that from which the sample measured paleomagnetically was obtained, some doubt remained. The actual sample on which the polarity was measured has now been dated, and it gives a result of 2.5 ± 0.2 m.y. This rock was somewhat altered. The date appears to be too young to be consistent with the stratigraphy; hence both dates should be treated with reservation until further detailed studies are made. They have not been included in Figure 3. Three samples from an alkali basalt lava flow of the Upper Waianae series exposed in a quarry at Kaena Point, Oahu, have reversed polarity [*Tarling*, 1965]. One of these samples has now been dated at 2.80 ± 0.06 m.y. The rock was quite unaltered and appears to be ideal for dating. Hence there are now two results for the Gauss normal epoch that have reversed polarity (Figure 3). The need for additional data in this epoch is obvious.

The dates of 3.55 and 3.9 m.y. (two localities) for Victorian basalts from lava flows that have reversed polarity are three more results in the Gilbert reversed epoch of *Cox et al.* [1964]. The Makaweli formation of Kauai, Hawaii, was dated by *McDougall* [1964] at between 3.8 and 4.0 m.y. *Cox et al.* [1965] reported reversed polarity from a number of samples collected independently from this formation. These data are also included in Figure 3.

Two results at 4.35 and 4.5 m.y. from Victoria are from lavas that have normal and reversed polarity, respectively. Only a single oriented sample (GA 1141), which has normal polarity, was collected from the Grange Burn locality. The meaning of these results must await further data.

The dating and paleomagnetic studies on the Newer Volcanics of Victoria provide new information on two of the polarity epoch boundaries and generally are consistent with previous results on polarity versus age. The data give additional support for the hypothesis that the geomagnetic field has changed its polarity at irregular intervals in the geological past. Together with the recent work on Réunion [*Chamalaun and McDougall*, 1966] this study is of significance because it is the first of its kind concerned with an extensive area in the southern hemisphere.

Addendum. Since this paper was submitted for publication *Doell and Dalrymple* [1966] have reported polarity determinations on rocks whose measured ages lie between 0.7 and 0.9 m.y. Their results are fully consistent with those reported in the present paper. The combined data provide a much clearer picture of the behavior of the geomagnetic field in this age range and suggest the following time scale: present to 0.70 m.y. ago, normal polarity; 0.70 to 0.88 m.y. ago, reversed polarity; 0.89 to 1.00 m.y. ago, normal polarity; 1.00 to 1.9 m.y. ago, reversed polarity.

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Appendix

GA 1064, GA 1172. Framlingham quarry, 4.2 km NNW of Panmure. Panmure 1:63,360 map, 632E, 761N. *Green and Irving* [1958] site NV 27. Alkali olivine basalt at least 15 m thick. Coarse-grained rock consisting of olivine phenocrysts in a groundmass of titanaugite, plagioclase, olivine, and iron oxide, intersertal finely crystalline feldspathic material forming 5 to 10% of the rock. This material is slightly altered. Normal polarity; age, 0.57 ± 0.03 m.y.

GA 1017, GA 1060, GA 1116. Albion quarry, Alphington, near Melbourne. Ringwood 1:50,000 map (no. 7922-111), 069E, 412N. *Green and Irving* [1958] site NV 15. Quarry is cut in a single olivine basalt lava at least 30 m thick. GA 1017 and GA 1116 from north end of quarry, below scoriaceous top of the lava flow. Both samples are unaltered olivine basalts with a well-developed fluidal texture. Less than 3% of finely crystalline feldspathic material occurs intersertally. GA 1060 from lowest point in quarry at its southeast corner, about 300 m southeast of GA 1017. Olivine basalt in which the main minerals, olivine, clinopyroxene, and plagioclase, show incipient alteration. Intersertally there occurs about 10% of very poorly crystallized to cryptocrystalline material which is partly chloritized. Reversed polarity; age, 0.81 ± 0.03 m.y.

GA 1061, GA 1171. Merri Creek quarry, Melbourne. Melbourne 1:63,360 map (no. 848, zone 7), 023E, 414N. *Green and Irving* [1958] site NV 12. Quarry cut in a single flow at least 30 m thick. GA 1061 about 20 m vertically below top of quarry, GA 1171 from floor of quarry. Olivine basalt, composed of fresh olivine phenocrysts in a groundmass of olivine, clinopyroxene, plagioclase, iron ore, and 10 to 15% of colorless to pale brown isotropic to slightly anisotropic glass in which microlites occur. Minor amygdaloidal opal. Reversed polarity; age, 2.20 ± 0.07 m.y.

GA 1023. Small disused quarry 1 km south of Dunnstown, 10 km ESE of Ballarat. Ballarat 1:63,360 map (no. 836, zone 7), 034E, 608N. Although these are the grid references given by *Green and Irving* [1958] for their site NV 18, petrographic examination of thin sections and paleomagnetic measurements made during this study indicated that this quarry was not sampled by them. See GA 2095 for the correct grid references for site NV 18. Massive olivine basalt which is holocrystalline and free of alteration except for some iddingsitization of olivine. Normal polarity; age, 2.53 ± 0.07 m.y.

GA 1021, GA 1115. Newport quarry, Melbourne. Melbourne 1:63,360 map, (no. 848, zone 7), 915°E, 332°N. *Green and Irving* [1958] site NV 3. The quarry is cut in a single olivine basalt lava which is about 15 m thick. GA 1021 from north wall of westernmost pit, and GA 1115 from north wall of easternmost pit. The lava is a slightly vesicular, coarse-grained subophitic basalt, which contains about 10% of reasonably well crystallized, intersertal, fine-grained feldspathic material. Normal polarity; age, 2.49 ± 0.15 m.y. (GA 1021), 2.50 ± 0.20 m.y. (GA 1115).

GA 1019. McGrath's quarry, Braybrook, near Melbourne. Melbourne 1:63,360 map (no. 848, zone 7), 891E, 415N. *Green and Irving* [1958] site NV 9. Appears to be a single lava flow at least 15 m thick. Massive, slightly vesicular rock, which contains about 5% of isotropic colorless glass speckled with small iron oxide granules. Normal polarity; age, 2.63 ± 0.15 m.y.

GA 1020, GA 1173. Albion quarry, Sunshine. Melbourne 1:63,360 map (no. 848, zone 7), 860E, 413N. *Green and Irving* [1958] site NV 10. GA 1020 from northern pit, which is cut into a somewhat vesicular lava about 10 m thick. Ophitic olivine basalt; the olivine is extensively altered to iddingsite. Intersertally there occurs about 10% of poorly crystallized feldspathic material. GA 1173 from north face of southern pit, adjacent to crushing plant. Difficult to determine whether this sample is from the same flow as GA 1020 or from an underlying flow. Coarse-grained olivine basalt; the olivine is altered at margins. Intersertally 10 to 15% of essentially isotropic, colorless to pale gray glass. Minor calcite in amygdules. Normal polarity; age, 2.55 ± 0.15 m.y. (GA 1020), 2.74 ± 0.20 m.y. (GA 1173).

GA 1066, GA 1174. Ararat quarry, western Victoria. Ballarat 1:253,440 map, 593E, 395N. *Green and Irving* [1958] sites NV 28, 29. The two adjacent pits are about 10 m deep and cut in a single, irregularly jointed, somewhat vesicular lava flow. Subophitic olivine basalt with about 10% of essentially isotropic glass containing a few microlites. Olivine is partly altered to iddingsite. Slight alteration of glass in GA 1066. Reversed polarity; age, 3.55 ± 0.10 m.y.

GA 2095. Operating quarry 1.4 km east of Dunnstown, 11 km ESE of Ballarat. Ballarat 1.63,360 map (no. 836, zone 7) 048E, 617N. *Green and Irving* [1958] site NV 18, although grid references given incorrectly in their paper. Massive fine-grained olivine basalt at least 10 m thick. Phenocrysts of olivine and plagioclase occur in a holocrystalline, fresh, fluidal groundmass consisting of plagioclase, pyroxene, iron oxide, and some iddingsitized olivine. Reversed polarity; age, 3.90 ± 0.10 m.y.

GA 1065. Menzel's quarry, Hamilton, western Victoria. Penhurst 1:63,360 map, 015E, 421N. *Green and Irving* [1958] site NV 30. Single lava flow at least 15 m thick. Holocrystalline olivine basalt which is free of alteration, except for iddingsitization of olivine. Reversed polarity; age, 3.91 ± 0.15 m.y.

GA 1141. Grange Burn, about 5 km west of Hamilton, western Victoria. See *Turnbull et al.* [1965]. Olivine basalt containing about 10 to 15% of pale to dark brown glass and cryptocrystalline material which is probably devitrified glass. Normal polarity; age, 4.35 ± 0.20 m.y.

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GA 1018, GA 1062. Fowler's quarry, off Keilor Road, Essendon, near Melbourne. Sunbury 1:63,360 map (no. 839, zone 7). 918E, 457N. *Green and Irving* [1958] site NV 11. The quarry appears to be cut into a single lava flow, of which about 30 m thickness is exposed. Samples from north end and south end of quarry, respectively. Massive, slightly vesicular olivine basalts. Unaltered rocks with a few per cent of pale brown isotropic glass and about 3% of green(?) opaline material in amygdules. Reversed polarity; age, 4.50 ± 0.15 m.y.

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